RAJALAKSHMI ENGINEERING COLLEGE

Akshaya S 251001013-2025-IT

A2

Started on	Tuesday, 25 August 2026, 6:28 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 6:29 PM
Time taken	53 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int a[], int low, int high, int key) {
    if (low > high)
        return -1;

    int mid = (low + high) / 2;

    if (a[mid] == key)
        return mid;
    else if (a[mid] > key)
        return binarySearch(a, low, mid - 1, key);
    else
        return binarySearch(a, mid + 1, high, key);
}

int findPair(int a[], int n, int x, int i) {
    if (i >= n)
        return 0;

    int j = binarySearch(a, i + 1, n - 1, x - a[i]);

    if (j != -1) {
        printf("%d\n%d", a[i], a[j]);
        return 1;
    }

    return findPair(a, n, x, i + 1);
}

int main() {
    int n, x;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    if (!findPair(a, n, x, 0))
        printf("No");

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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